Varendra University

Section	All

Department of Computer Science and Engineering
6th Semester Midterm Examination (Spring-2026)
Course Code: CSE 3207
Course Title: Digital Signal Processing

Time: 1.50 hours **Marks: 24** (3x8)

(Answer any three of the following questions, including question No. 01)
The figures in the right margin indicate full marks for the respective question.
All parts of each question must be answered sequentially.

Question no	*CO's no.*	*Description*	**Taxonomy domain/level**	**POs**	**WK**	**WP**	**EA**
1	*CO1*	*Describe the concepts of signal and its discretization process*	*Cognitive/ Understand*	*PO1*	*WK2*	-	-

1. a) **Illustrate** the statement "A discrete-time sinusoid is periodic only if its frequency f is a rational number." **2**

   b) **Explain** the meaning of the rate of oscillation in a discrete-time sinusoid. **2**

   c) Two analog sinusoidal signals are given as: **4**

$$x_1(t) = \cos 2\pi\,(50)t$$
$$x_2(t) = \cos 2\pi\,(10)t$$

   Both signals are sampled with a sampling frequency of F_s= 40 Hz.

   Interpret the discrete-time signals $x_1(n), x_2(n)$ derived from the given signal.

2. a) What is meant by sampling, quantization, and coding in digital signal processing? **3**

   b) State the Sampling Theorem. **2**

   c) List the differences between continuous and discrete signals. **3**

3. a) Determine the power and energy of the unit step sequence **3**

$$u(n) \equiv \begin{cases} 1, & \text{for } n \geq 0 \\ 0, & \text{for } n < 0 \end{cases}$$

   b) A signal $x(n)$ is graphically illustrated in Figure 1. Show the graphical representations of the signals $x(n-2)$ and $x(n+1)$. **3**

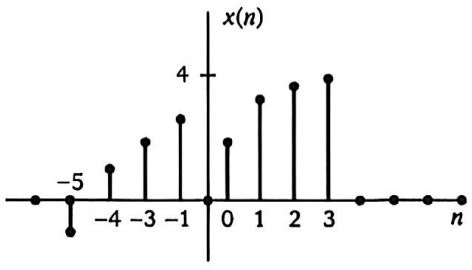


Figure 1: Graphical Representation of the Discrete-Time Signal $x(n)$

c) Determine the outputs of the following systems to the input signal **2**

$$x(n) = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & otherwise \end{cases}$$

i. $y(n) = x(n-1)$

ii. $y(n) = x(n+1)$

4. a) A system is described as $y(n) = x(n) - x(n-1)$. Analyze whether this system is time-invariant or not. **4**

b) Determine the output $y(n)$ of a relaxed linear time-invariant system with impulse response **4**

$$h(n) = b^n u(n), \qquad |b| < 1$$

when the input is a unit step sequence, that is,

$$x(n) = u(n).$$

Varendra University

Department of Computer Science and Engineering

6th Semester Final Examination (Spring-2026)

Course Code: CSE 3207

Course Title: Digital Signal Processing

Time: 2 hours **Marks: (4x9) =36**

(Answer any four of the following questions, of which Question no.1 is mandatory)

The figures in the right margin indicate full marks for the respective question.

All parts of each question must be answered sequentially.

COs	Description	Taxonomy domain/level	POs	WK	WP	EA
CO3	Apply Fourier transform techniques to analyze and implement frequency domain signal processing and filtering.	Cognitive/ Apply	3	5	-	-

1. a) **Calculate** the 4-point DFT of the sequence $x(n) = \{0, 1, 2, 3\}$ using the radix-2 DIT FFT algorithm. Draw the complete butterfly diagram showing all stages and state the twiddle factors W_2^0, W_4^0, W_4^1 used at each stage. **4.5**

b) **Illustrate** the 4-point DIF-FFT butterfly flow graph and use it to compute the DFT of **4.5**

$$x[n] = \cos\left(\frac{n\pi}{2}\right) \text{ for } 0 \leq n \leq 3.$$

2. a) Using the radix-2 Decimation-in-Time (DIT) FFT algorithm, draw the butterfly diagram for the following 8-point sequence: **4.5**

$$x[n] = \{1,0,0,0,0,0,0,0\}, \quad 0 \leq n \leq 7$$

Hence, compute the 8-point FFT of the sequence.

b) Find the IDFT of the sequence **4.5**

$$X(k) = \{10, -2 + j2, -2, -2 - j2\}$$

using DIT algorithm.

3. a) Explain why the Fourier Transform is important in digital signal processing. Provide one real-life example to support your explanation. **3**

b) List the differences between DIT FFT and DIF FFT. **3**

c) Compute the N-point DFT of the following sequences for $0 \leq n \leq N - 1$: **3**

$$x[n] = \delta[n].$$

4. a) Express the twiddle factor matrix for an 8-point sequence. **4**

b) Compute 8-point DFT of $x[n] = \{1, -1, 1, -1, 0, 0, 0, 0\}$. **5**

5. a) State and prove the periodicity and linearity properties of the Discrete Fourier Transform (DFT). **4**

b) Compute IDFT of the following function with N = 4: **5**

$$X(k) = \{6, (-2 + j2), -2, (-2 - j2)\}$$

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Q1. Given $x[n] = \{1, 2, -3, 0, 4, -1\}$, arrow at -3. Sketch $x[n]\{u[n+1] - u[n-4]\}$. **[10 Marks]**

Q2. Determine whether $x[n] = \cos(2\pi n/7)$ is periodic. If periodic, find N_0. **[10 Marks]**